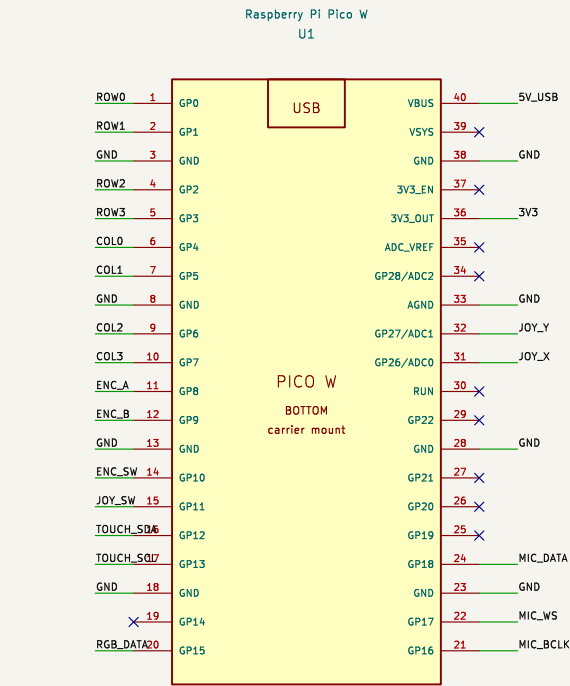


MACROPAD / 13-key USB + wireless-capable carrier

REV A – engineering prototype | 3.3 V GPIO | USB-only power | KiCad 9 editable source

01 / CONTROLLER



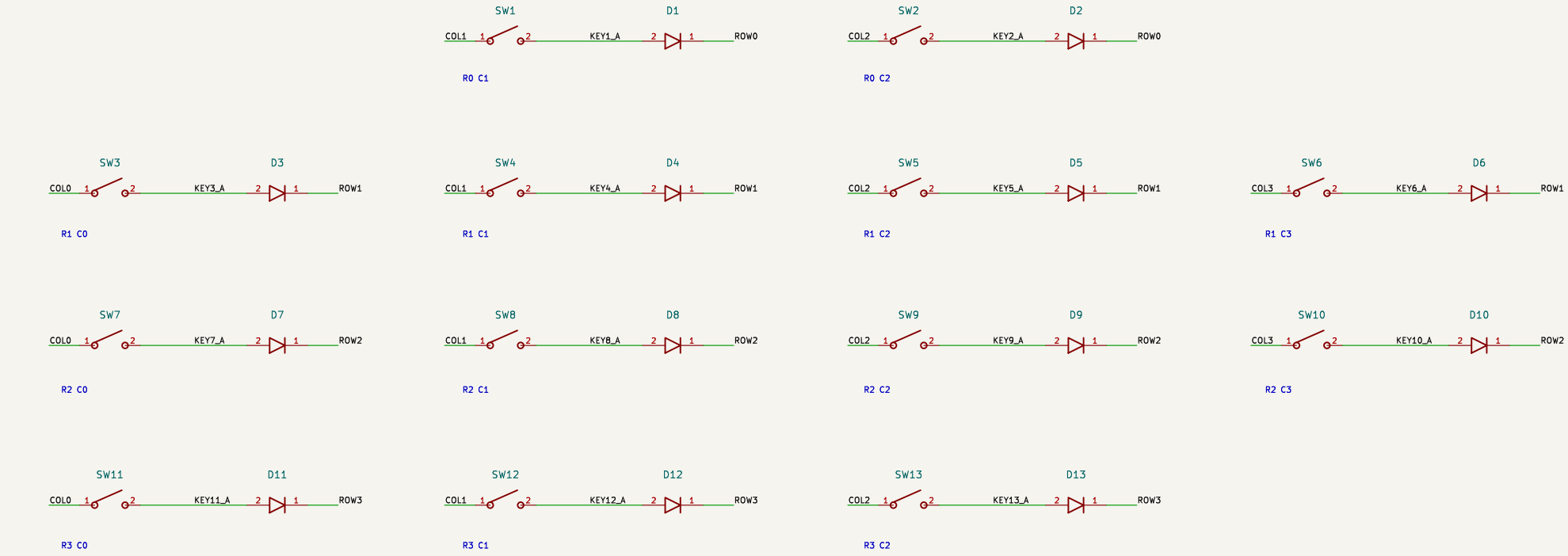
U1 solders to bottom carrier castellations

Keep copper / metal clear of antenna end

Unused Pico pins explicitly NC

02 / 13-KEY DIODE MATRIX

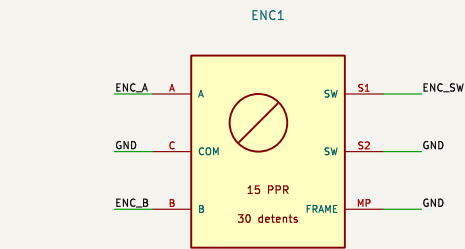
COL -> switch -> A diode K -> ROW; drive one ROW low, read COL pull-ups



D1–D13: 1N4148W / SOD-123, pin 1 = cathode stripe

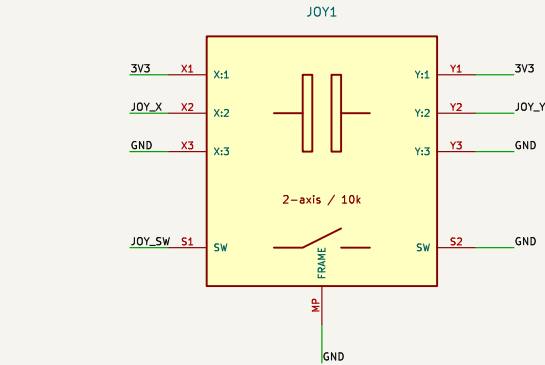
SW1–SW13: Kailh sockets underneath; switches inserted from top

03 / ROTARY CONTROL



EC11E15244B2 | internal GPIO pull-ups

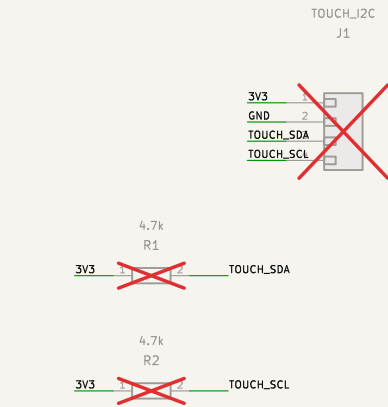
04 / ANALOG JOYSTICK



RKJXV1224005 | 2 x 10k | 3.3 V only

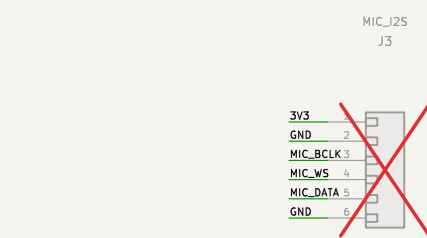
Axis directions calibrated in firmware

05 / OPTIONAL I2C TOUCH



DNP R1/R2 if module has pull-ups

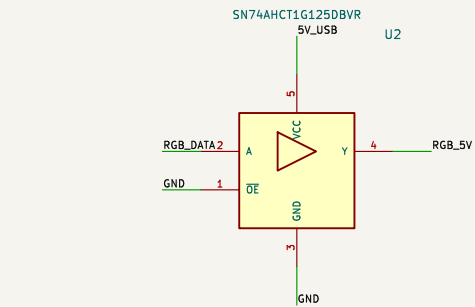
06 / I2S MICROPHONE RESERVE



3.3 V module only; no analog microphone

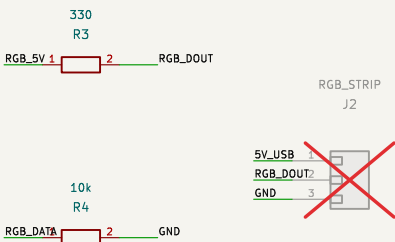
GP16 BCLK / GP17 WS / GP18 DATA

07 / 5 V RGB OUTPUT



U2 TTL threshold accepts 3.3 V; /OE tied low

Six external WS2812B LEDs maximum; limit brightness/current in firmware



C2 optional; check USB Inrush; external strip not Included

All optional headers / R1–R2 / C2 marked DNP

Check footprint drawings + physical first article before manufacture

